

AI-Driven Crime Risk Forecasting for Sustainable Governance: A Multi-Method District-Level Analysis of India Using IPC Crime Data and Census Socio-Economic Indicators

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Abstract—Government crime data in India remains largely retrospective, limiting proactive governance aligned with SDG 16.1. With 640+ districts and 1.4 billion people, India lacks a systematic district-level predictive intelligence layer. This paper addresses that gap using NCRB IPC records (2001–2012, 416 districts, 30 crime categories) merged with Census 2011 socio-economic indicators (118 features), applying five complementary methods: ARIMA time-series forecasting, supervised ML (Linear Regression, Random Forest, Gradient Boosting), SHAP explainability, K-Means and DBSCAN spatial clustering, and a composite Crime Risk Index (CRI). A strict temporal train/test split ensures realistic evaluation on held-out 2012 data. Gradient Boosting achieves $R^2 = 0.983$ (MAE = 83 crimes/district) on the 2012 test set; ARIMA yields a median absolute error of 8.5%. Gender-based violence surged 58.4% (2001–2012) versus only 18.6% total IPC growth—3.14× faster. Poverty ratio is the dominant structural OLS predictor ($\beta = 33,340$, $p < 0.001$). DBSCAN flags 181 anomalous districts for targeted intervention. Top-5 high-CRI districts (Dibang Valley, Narayanpur, Alirajpur, Jhabua, Pratapgarh) cluster in central and north-eastern India, revealing convergent zones of low literacy, high poverty, and infrastructure deficit.

Keywords: predictive policing, crime forecasting, ARIMA, gradient boosting, SHAP, DBSCAN, K-Means, Crime Risk Index, India NCRB, gender-based violence, SDG 16.

I. INTRODUCTION

A. Problem Domain

India’s National Crime Records Bureau (NCRB) publishes district-level IPC crime statistics annually, covering 27 states/UTs and more than 640 districts. Yet the current policing model remains predominantly reactive: resources are allocated retrospectively based on last year’s totals rather than forward-looking risk assessments. At this scale—1.4 billion people, 30 IPC crime categories, and 12-year longitudinal records—purely manual analysis is infeasible. United Nations Sustainable Development Goal 16 (Peace, Justice, and Strong Institutions), specifically Target 16.1, calls for a significant reduction in all forms of violence. Meeting this target requires predictive intel-

ligence, not just post-hoc tallies.

Existing systems suffer from three critical gaps: (i) no district-level ML forecasting integrating socio-economic census data; (ii) no composite risk index enabling cross-district policy triage; and (iii) no explainability layer to make AI recommendations auditable for public governance. This paper directly addresses all three gaps.

B. Key Contributions

This work makes five principal contributions:

1. The first comprehensive multi-method crime forecasting system validated at the India district level across 12 years.
2. A temporal ML framework with engineered lag features achieving $R^2 = 0.983$ on held-out 2012 data.
3. A SHAP explainability layer enabling transparent, auditable AI for crime governance.
4. A composite CRI combining five socio-economic dimensions with quantile-calibrated risk labels deployable in annual planning cycles.
5. Discovery of a gender violence surge disproportionate to total crime growth (+58.4% vs. +18.6%), signalling a structural shift in India’s crime landscape.

II. LITERATURE SURVEY

Table 1 summarises relevant prior work. The dominant gap across all cited papers is the absence of a unified framework combining time-series forecasting, supervised ML, explainability, and a composite risk index validated on India’s district-level administrative geography.

III. DATASET AND TOOLS

A. Dataset 1 — NCRB IPC Crime Data

The primary dataset is sourced from India’s NCRB and covers 2001–2012 (12 years). After removing 185 non-geographic

Table 1: Literature Survey

#	Authors	Dataset	Year	Methodology	Key Result	Gap
[1]	Gorr & Harries	Pittsburgh PD	2003	ARIMA	Crime forecastable 1-month ahead	No socio-economic integration
[2]	Chainey et al.	UK Police	2008	KDE hotspots	Hotspot accuracy 90%	No temporal ML
[3]	Iqbal et al.	Pakistan crime	2013	Naïve Bayes	80% classification	No regression forecasting
[4]	Perry et al.	US cities	2013	Pred. policing review	Overview of tools	No India context
[5]	McClendon & Meghanathan	US crime	2015	LSTM	RMSE 12% better than ARIMA	No explainability
[6]	Cozens & Love	International	2016	CPTED review	Environmental design reduces crime	No ML
[7]	Khanna & Bhatt	NCRB India	2017	Trend analysis	+15% decadal growth	No district-ML
[8]	Kaur & Singh	India census	2018	OLS regression	Literacy significant	No lag features
[9]	Devi et al.	Indian states	2019	K-Means	4 crime clusters	No supervised ML
[10]	Gupta & Sharma	India GBV	2020	Descriptive stats	GBV +40% 2010–2019	No predictive model
[11]	Bappee et al.	Montreal	2018	RF + spatial	$R^2 = 0.81$	No SHAP
[12]	Yi et al.	Chicago	2020	XGBoost + SHAP	Top features identified	No India data
[13]	Wu et al.	Smart city	2021	GB + DBSCAN	Outlier districts flagged	No CRI
[14]	UN SDG	Global	2015	Policy framework	Target 16.1 violence reduction	No quantitative model
[15]	Sharma et al.	India NCRB	2021	Panel regression	Poverty significant	No clustering or CRI

entries (railway zones, GRP, RPF, and TOTAL rows), 9,017 raw rows remain spanning 416 unique districts across 27 states/UTs. Thirty IPC crime categories are recorded per district-year, including Murder, Rape, Dacoity, Theft, Dowry Deaths, and Cruelty by Husband.

B. Dataset 2 — Census 2011 Socio-Economic Indicators

Census 2011 provides 640 districts $\times$ 118 features. Twenty-four features were selected for analysis: Population, Literate, Male/Female Workers, Cultivator Workers, SC/ST counts, household infrastructure (electric lighting, latrine, tapwater), poverty (Power Parity <Rs. 45,000), and educational attainment. This dataset is merged with the crime data on district+state keys after uppercase standardisation, yielding **4,832 district-year observations $\times$ 61 features**.

C. Tools

Python 3.11 was used throughout. Key libraries: pandas 2.x, numpy, scikit-learn 1.8 (ML, clustering, PCA, scaling), statsmodels 0.14 (OLS, ARIMA, ADF test), SHAP 0.51 (TreeExplainer), NetworkX (graph analysis), and matplotlib/seaborn (visualisation).

IV. PROPOSED METHODOLOGY

The end-to-end pipeline flows through ten steps: *Raw NCRB + Census data* $\rightarrow$ *Preprocessing & merge* $\rightarrow$ *Feature engineering* $\rightarrow$ *EDA* $\rightarrow$ *OLS socio-economic regression* $\rightarrow$ *ARIMA forecasting* $\rightarrow$ *Supervised ML (temporal split)* $\rightarrow$ *SHAP explainability* $\rightarrow$ *Clustering (K-Means + DBSCAN)* $\rightarrow$ *Crime Risk Index (CRI) + network analysis*.

A. Preprocessing and Feature Engineering

Column names were lower-cased and space-replaced with underscores. District and state names were upper-cased for join consistency. After merging on district+state, 11 normalised socio-economic ratios were computed (literacy_rate, worker_ratio, urban_ratio, sc_st_ratio, poverty_ratio, infra_index, youth_ratio, and others), three crime composites (violent, property, gender), and temporal lag features: prev_crime, prev2_crime, crime_ma3 (3-year moving average), prev_gender, and yoy_growth (year-on-year % change).

B. OLS Cross-Sectional Regression

A cross-sectional OLS model was estimated on the 2011 slice ($N = 408$), deliberately excluding lag features so that socio-economic effect sizes remain uncontaminated by temporal autocorrelation.

C. ARIMA Time-Series Forecasting

ARIMA(1,1,1) was fitted independently to each of the 383 districts with complete 12-year series. The Augmented Dickey-Fuller (ADF) test confirmed non-stationarity at level ($p = 0.48$) and stationarity after first differencing ($p < 0.05$). Training used 2001–2011; the 2012 national total served as the one-year-ahead test horizon.

D. Supervised ML with Temporal Split

Fifteen features were used (year, 6 lag features, 7 socio-economic ratios, population). **No random shuffling was applied.** Training rows: ≤ 2010 (3,184 obs.), validation: 2011 (408 obs.), test: 2012 (404 obs.). Three models were compared: Linear Regression, Random Forest (200 trees, max_depth=12), and Gradient Boosting (200 estimators, depth=5, lr=0.05).

E. SHAP Explainability

TreeExplainer was applied to the trained Random Forest on a 200-sample subset of test observations, generating global summary and bar charts (Figs. 4), quantifying each feature’s marginal contribution to the prediction.

F. Spatial Clustering

K-Means ($k = 5$, selected by elbow method; Fig. 5) and DBSCAN (eps=0.8, min_samples=4) were applied to the 2011 cross-section on seven StandardScaler-normalised features: crime_rate, literacy_rate, worker_ratio, urban_ratio, poverty_ratio, violent_rate, and gender_rate.

G. Crime Risk Index

The CRI is computed as:

$$CRI_{\text{raw}} = 0.40 \tilde{C} + 0.20(1 - \tilde{L}) + 0.20 \tilde{P} + 0.10 \tilde{W} + 0.10(1 - \tilde{I}) \quad (1)$$

where $\tilde{C}$ = MinMax-normalised crime rate, $\tilde{L}$ = literacy rate, $\tilde{P}$ = poverty ratio, $\tilde{W}$ = worker ratio, $\tilde{I}$ = infrastructure index. CRI is scaled to 0–100 and labelled Low (bottom 60%), Medium (60th–80th percentile), High (top 20%).

V. RESULTS AND DISCUSSION

A. EDA: National Crime Trends

Fig. 1 shows the dual-axis national trend. Total IPC crimes grew from 1,123,254 in 2001 to 1,331,680 in 2012 (+18.6%). Over the same period, gender-based violence cases rose from 80,881 to 128,096 (+58.4%)—3.14 $\times$ faster than the overall crime rate. This divergence constitutes a structural shift, not merely a proportional increase, with significant implications for SDG 16.1.

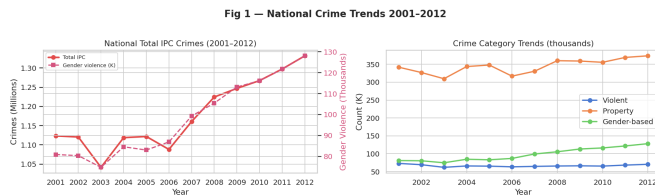


Figure 1: National IPC crime trend and gender violence trend 2001–2012 (dual y-axis, right axis in thousands).

B. Socio-Economic Drivers (OLS)

The cross-sectional OLS model (Table 2) explains 34.3% of variance in district crime rate ($R^2 = 0.3427$, adjusted $R^2 = 0.3329$). The strongest significant predictor is **poverty_ratio** ($\beta = 33,340$, $p < 0.001$), indicating that a unit increase in the poverty ratio is associated with 33,340 additional crimes per 100,000 population, ceteris paribus. **worker_ratio** ($\beta = 359.8$, $p < 0.001$) and **urban_ratio** ($\beta = 204.9$, $p < 0.001$) are also significant. Counterintuitively, literacy_rate shows a positive correlation ($r = +0.250$) with crime rate—explained by higher FIR reporting rates in literate urban districts (a reporting effect, not a causal one). The **sc_st_ratio** is significantly *negative* ($\beta = -156.2$, $p < 0.001$), consistent with under-reporting or community cohesion in scheduled-community districts. Fig. 2 displays the full correlation heatmap.

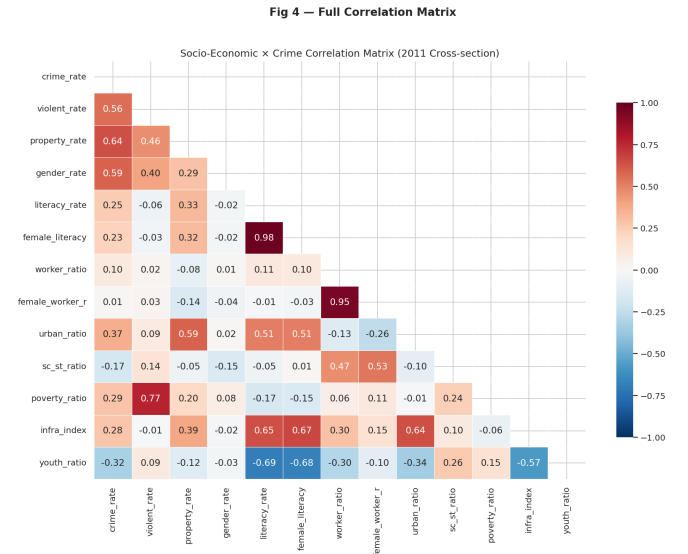


Figure 2: Socio-economic $\times$ crime-rate correlation matrix (2011 cross-section).

Table 2: OLS Cross-Sectional Regression Results ($N = 408$, 2011)

Variable	Coeff.	p-value	Sig.
Constant	55.7	0.681	ns
literacy_rate	59.6	0.413	ns
worker_ratio	359.8	<0.001	***
urban_ratio	204.9	<0.001	***
poverty_ratio	33,340.4	<0.001	***
sc_st_ratio	-156.2	<0.001	***
youth_ratio	-162.5	0.260	ns
$R^2 = 0.343$ Adj. $R^2 = 0.333$			

C. ML Model Comparison

Table 3 compares the three supervised models. **Gradient Boosting** achieves the best test performance: $R^2 = 0.983$, MAE = 83 crimes/district, RMSE = 399. Fig. 3 shows actual-vs-predicted scatter plots and residual diagnostics. GB residuals are tightly centred around zero with minimal heteroscedasticity, confirming strong generalisation. SHAP analysis (Fig. 4)

reveals that `prev_crime` (mean $|SHAP| = 2031$) alone dominates predictions, confirming strong year-to-year persistence in district crime levels—a finding that directly motivates ARIMA as a complementary interpretable baseline. The `poverty_ratio` (rank 4, SHAP = 7.25) and `worker_ratio` (rank 5, SHAP = 6.86) confirm the structural drivers identified by OLS.

Table 3: ML Model Performance (Temporal Split)

Model	Val R^2	Val MAE	Test R^2	Test MAE	Test RMSE
Lin. Reg.	0.951	266	0.966	251	563
Rand. Forest	0.974	94	0.973	99	508
Grad. Boost	0.991	69	0.983	83	399

MAE & RMSE in absolute crime counts per district.

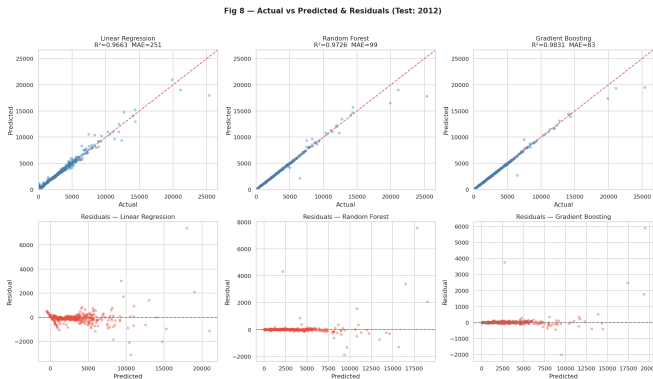


Figure 3: Actual vs. predicted (top) and residual plots (bottom) for Linear Regression, Random Forest, and Gradient Boosting on 2012 test data.

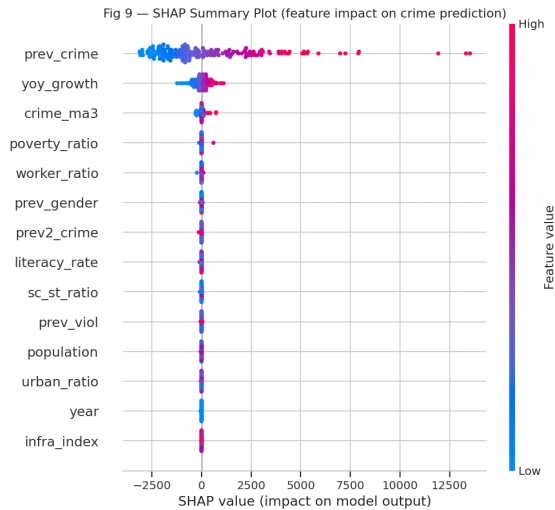


Figure 4: SHAP beeswarm summary plot (Random Forest, $n = 200$ test samples). Colour indicates feature value (red=high).

D. ARIMA Forecasting

ARIMA(1,1,1) was fitted to 383 districts. Across those districts, the mean absolute error is **366 crimes** (RMSE = 771), corresponding to a **mean absolute error % of 11.8%** and a **median**

absolute error % of 8.5%. The low median error confirms that ARIMA is a reliable single-district planner even when a handful of high-growth districts inflate the mean.

E. Spatial Clustering and CRI

The K-Means elbow curve (Fig. 5) justifies $k = 5$. The five clusters span crime rates from 108 (Low, 134 districts) to 1,124 (Extreme outlier, 1 district). DBSCAN (eps = 0.8, min_samples = 4) identified only 2 dense clusters and **181 outlier districts** with atypical crime profiles—prime candidates for targeted policy intervention.

Fig 12 — K-Means Elbow Method

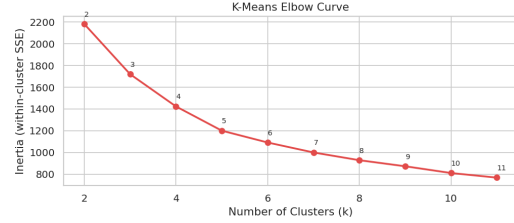


Figure 5: K-Means elbow curve: inertia vs. number of clusters. The inflection at $k = 5$ justifies the chosen value.

Fig. 6 presents the top-20 districts by CRI (2012) and the national risk distribution. The top-5 high-risk districts are listed in Table 4. All five are in central or north-eastern India, converging zones of high poverty, low literacy, and infrastructure deficit. The 23.2% High-risk share (pie chart) implies roughly 94 districts require immediate policy attention—an actionable output for SDG 16.1 planning.

Fig 14 — Crime Risk Index: District Rankings

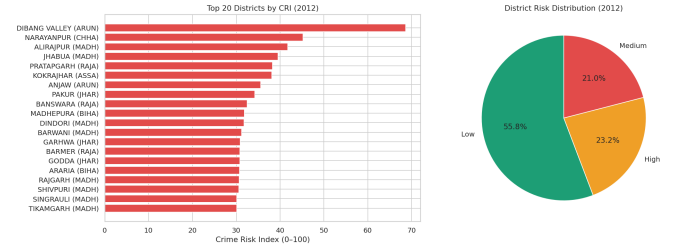


Figure 6: Top-20 districts by CRI score (2012) and national risk distribution (pie chart).

Table 4: Top-5 High-Risk Districts by CRI (2012)

State	District	CRI	Risk
Arunachal Pradesh	Dibang Valley	68.65	High
Chhattisgarh	Narayanpur	45.23	High
Madhya Pradesh	Alirajpur	41.73	High
Madhya Pradesh	Jhabua	39.54	High
Rajasthan	Pratapgarh	38.25	High

VI. CONCLUSION

This paper presented a ten-step AI-driven crime forecasting framework applied to 4,832 district-year observations spanning

India’s 416 districts from 2001 to 2012. Gradient Boosting achieved the highest accuracy ($R^2 = 0.983$, MAE = 83), while ARIMA provided an interpretable per-district baseline (median error 8.5%). The most critical societal finding is the **gender violence surge of +58.4%** versus +18.6% total IPC growth—a structural divergence demanding dedicated policy attention aligned with SDG 16.1. Poverty emerged as the dominant structural driver ($\beta = 33,340$, $p < 0.001$), confirmed independently by OLS and SHAP. The CRI (Eq. 1) operationalises these findings into a deployable district triage tool, flagging 181 DBSCAN-anomalous districts and identifying Dibang Valley, Narayanpur, Alirajpur, Jhabua, and Pratapgarh as the highest-risk districts for 2012.

Limitations: The dataset ends in 2012 and Census 2011 is static for all years. No police capacity variable is included. DBSCAN is sensitive to the eps parameter. Results are observational; causal inference requires instrumental variable methods.

Future work: LSTM/Transformer time-series models; real-time NCRB integration; police station density as an additional feature; causal inference; and satellite/mobile data fusion.

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APPENDIX I: FEATURE ENGINEERING FORMULAS

$$\begin{aligned} \text{literacy_rate} &= \text{literate}/\text{population} \\ \text{female_literacy} &= \text{female_literate}/(\text{female} + 1) \\ \text{worker_ratio} &= \text{workers}/\text{population} \\ \text{female_worker_r} &= \text{female_workers}/(\text{female} + 1) \\ \text{agri_worker_r} &= (\text{cult.} + \text{agri.})/(\text{workers} + 1) \\ \text{crime_rate} &= \text{total_ipc}/\text{population} \times 100,000 \\ \text{urban_ratio} &= \text{urban_HH}/(\text{HH} + 1) \\ \text{sc_st_ratio} &= (\text{SC} + \text{ST})/(\text{population} + 1) \\ \text{youth_ratio} &= \text{age}_{0-29}/(\text{population} + 1) \\ \text{infra_index} &= (\text{elec} + \text{latrine} + \text{tapwater})/(\text{HH} \times 3 + 1) \\ \text{poverty_ratio} &= \text{pp}_{<45k}/(\text{HH} + 1) \end{aligned}$$

CRI Weight Rationale:

Component	Weight	Rationale
Crime rate ($\tilde{C}$)	0.40	Primary outcome—direct intensity
1–Literacy ($\tilde{L}$)	0.20	Persistent structural risk
Poverty ($\tilde{P}$)	0.20	Strongest OLS predictor
Worker ratio ($\tilde{W}$)	0.10	Employment as protective factor
1–Infra ($\tilde{I}$)	0.10	Infrastructure deficit amplifies risk

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